



# Some Useful Tips And Tricks For Designing Drones

Choosing appropriate components is crucial to building a drone with desired specifications

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**U**nmanned aerial vehicles (UAVs), commonly known as drones, with their ability to access remote locations, have quickly progressed from military to numerous non-military sectors (such as in aerial or video photography). Drones can be customised for such specific applications as high-temperature resistance to act as first responders in case of a fire.

In this article we discuss some useful pointers that could help you save time and decrease cost if you are planning to build a custom UAV of your own.

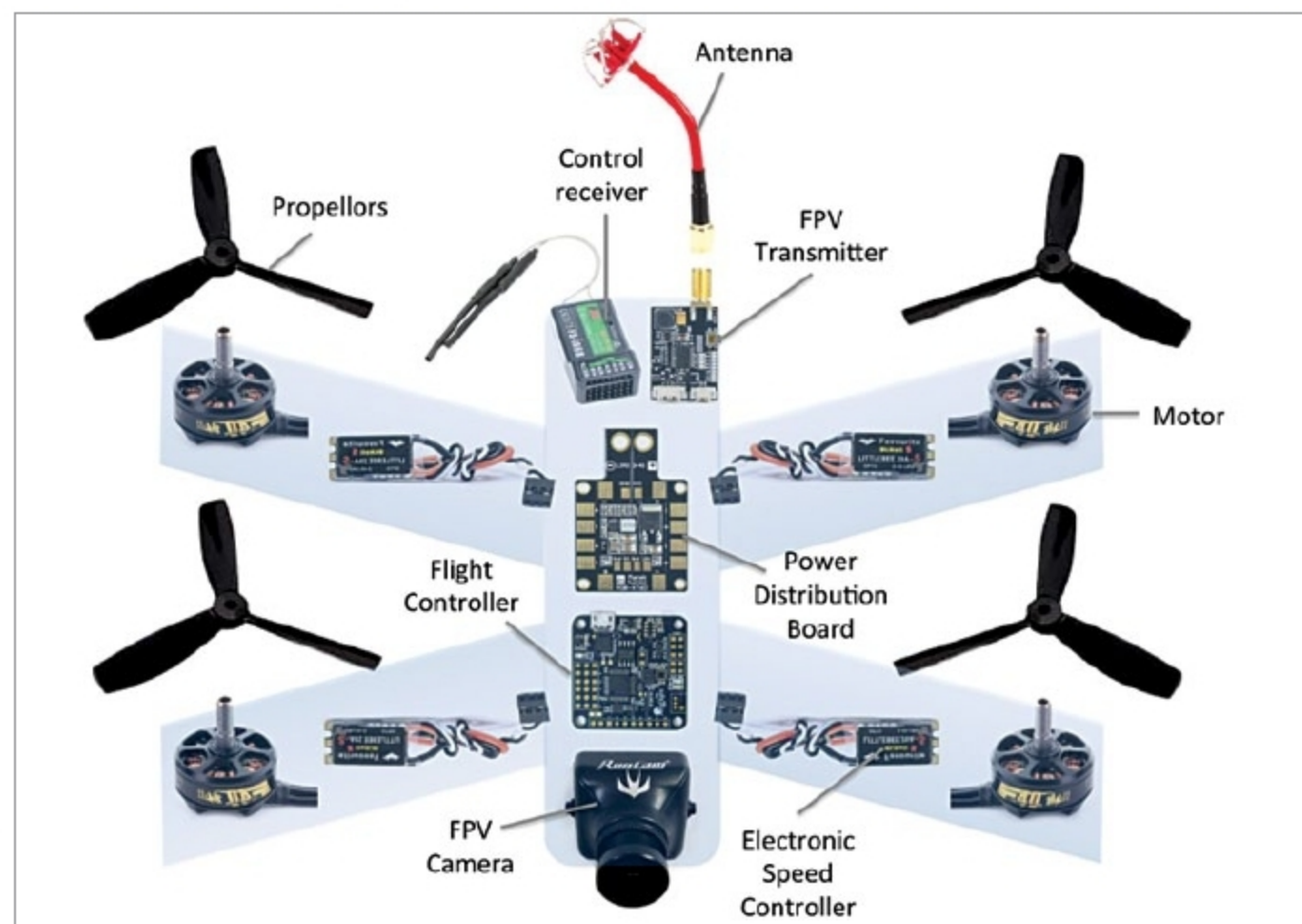
## Understanding the science and terminology

Educating yourself is of foremost importance, especially if you want to reduce your chances of redesigning. This includes

learning how flying becomes possible, different associated terminologies, and what impact do various components, centre of gravity and weight have on flight of the drone. Besides, many drone development platforms and kits are also available commercially to assist you.

In simple terms, four forces act on any aircraft while flying: weight, lift, drag and thrust. Weight, due to gravitational force, needs to be balanced by lift for an object to fly or at least remain afloat. If pressure at the top is lower than pressure at the bottom, the aircraft moves in an upwards direction. The drone is therefore designed to be lightweight.

Thrust or forward velocity depends on the shape of various parts like wings. Drag force slows down the aircraft by acting in the opposite direction of motion.



Standard components of a common quadcopter drone (Credit: <https://fpvdroneviews.com>)